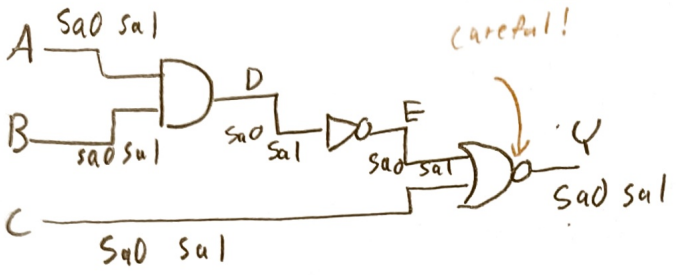
Fault Equivalence (Y is NOR)



Fault List

① A-Sa0 ~~A-Sa0~~
A-Sa1 ✓
① B-Sa0 ~~B-Sa0~~
B-Sa1 ✓ ✓
C-Sa0
④ C-Sa1 ~~C-Sa1~~
② ① D-Sa0 = A-Sa0, B-Sa0 ~~D-Sa0 = A-Sa0, B-Sa0~~
③ D-Sa1 ~~D-Sa1~~
③ E-Sa0 = D-Sa1
④ ② E-Sa1 = D-Sa0, A-Sa0, B-Sa0 ~~E-Sa1 = D-Sa0, A-Sa0, B-Sa0~~
④ Y-Sa0 = E-Sa1, DSa0, ASa0, BSa0, C-Sa1
Y-Sa1

Equivalent Faults

AND → Sa0
OR → Sa1
Nand → Sa0; Sa1
NOR → Sa1; Sa0
Not → Sa1; Sa0
Sa0, Sa1

Forward along Branches

Collapsing from Input to Output:

① A-Sa0 = B-Sa0 = D-Sa0 → D-Sa0 = A-Sa0, B-Sa0

② D-Sa0 = E-Sa1 → E-Sa1 = D-Sa0, A-Sa0, B-Sa0

③ D-Sa1 = E-Sa0 → E-Sa0 = D-Sa1

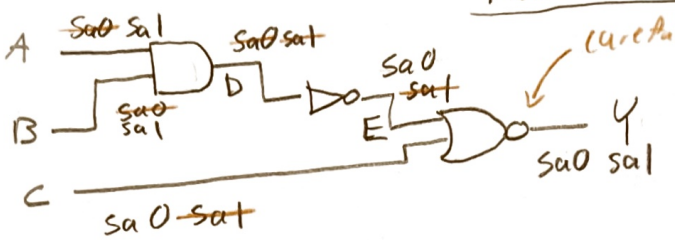
④ Y-Sa0 = E-Sa1, C-Sa1 → Y-Sa0 = E-Sa1, DSa0, ASa0, BSa0, C Sa1
this is what changes if last gate is NOR

Collapsed Fault List

A-Sa1
B-Sa1
C-Sa0
E-Sa0 = DSa1
Y-Sa0 = E-Sa1, D-Sa0, A-Sa0, B-Sa0, C-Sa1
Y-Sa1

Collapse Ratio: 6/12 = .5

Fault Dominance (Y is NOR)



After equivalence collapse:

A-Sa1

B-Sa1

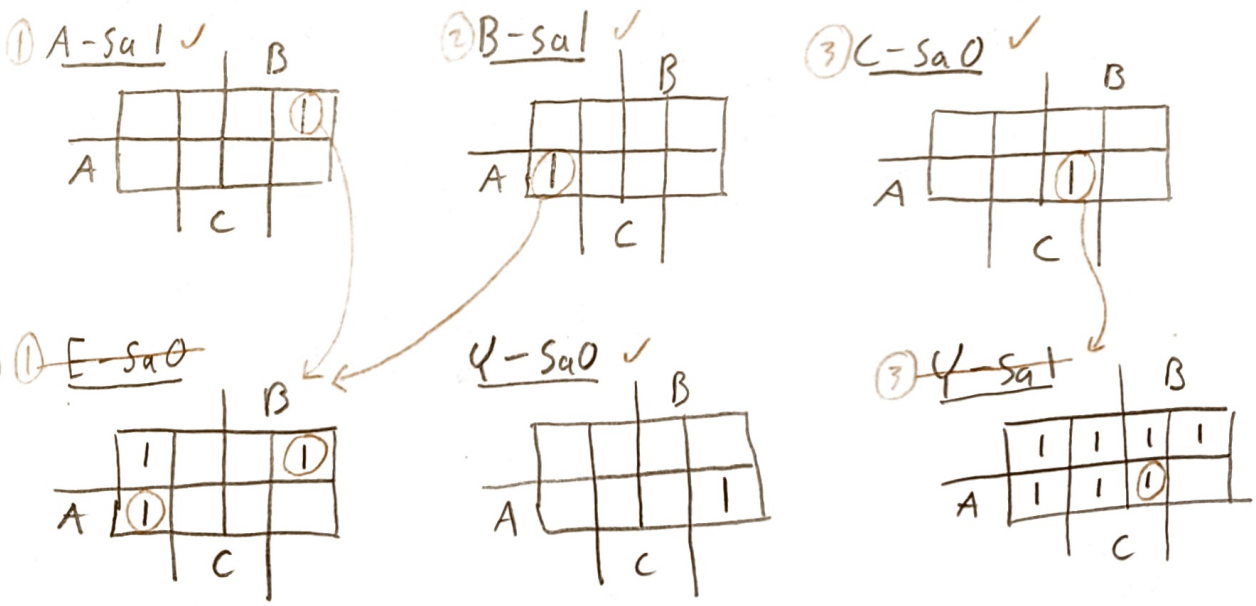
C-Sa0

E-Sa0 = D Sa1 → Dominated by A sa1 + B sa1

Y-Sa0 = E-Sa1, D-Sa0, A-Sa0, B-Sa0, C-Sa1

Y-Sa1 → [Dominated by ... Sa0, ...]

So we want to look @ all our tests and see which cover eachother. I'll do this exhaustively...



1 Any test for A-Sa1 also tests E Sa0 ↘ so we can remove E-Sa0

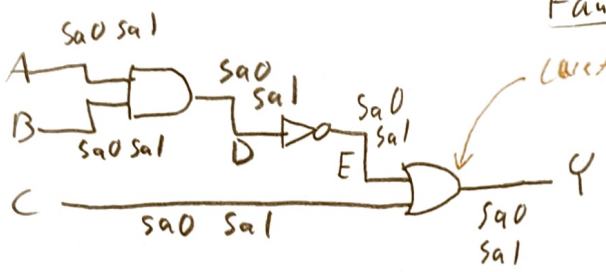
2 Any test for B-Sa1 also tests E Sa0 ↗

3 Any test for C-Sa0 also tests Y-Sa1 - so we can remove Y-Sa1

Final Fault List: A-Sa1, B-Sa1, C-Sa0, Y-Sa0

Collapse Ratio = 4/12 = .33

Fault Equivalence (Y is O.R.)



Equivalent Faults

And → sa0 all equal
OR → sa1 all equal
Nand → input sa0 → output sa1
NOR → input sa1 → output sa0
Not → in sa1 → out sa0
in sa0 → out sa1
Forward along Branches

Fault List

① ~~A-sa0~~
A-sa1
① ~~B-sa0~~
B-sa1
C-sa0
④ ~~E-sa1~~
② ① ~~D-sa0 = Asa0, Bsa0~~
D-sa1
③ E-sa0 = D-sa1
④ ② ~~E-sa1 = Dsa0, Asa0, Bsa0~~
Y-sa0
④ Y-sa1 = Esa1, Dsa0, Asa0, Bsa0, Csa1

Collapsing from Input to output

① Asa0 = Bsa0 = Dsa0 → D-sa0 = Asa0, Bsa0

② D-sa0 = Esa1 → E-sa1 = Dsa0, Asa0, Bsa0

③ D-sa1 = Esa0 → E-sa0 = Dsa1

④ Y-sa1 = E-sa1, C-sa1 → Y-sa1 = E-sa1, D-sa0, Asa0, Bsa0, C-sa1

* This is what changes if last gate is OR

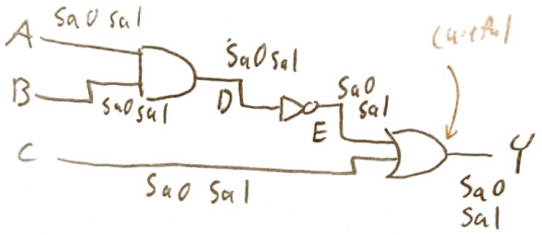
Collapsed Fault List

A-sa1
B-sa1
C-sa0
E-sa0 = D-sa1
Y-sa0
Y-sa1 = E-sa1, D-sa0, A-sa0, B-sa0, C-sa1

Collapse Ratio

6/12 = .5

Fault Dominance (Y is OR)



After Equivalence Collapse:

A-Sa1

B-Sa1

C-Sa0

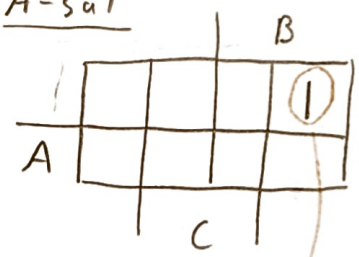
② ① E-Sa0 = D Sa1 → dominated by Asa1 + Bsa1

③ Y-Sa0 → dominated by C-Sa0

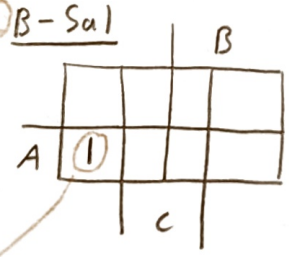
Y-Sa1 = E-Sa1, D Sa0, A-Sa0, B-Sa0, C-Sa1

So now we we want to look @ all tests and see which cover eachother. We do this exhasively

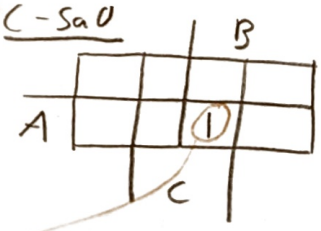
① A-Sa1



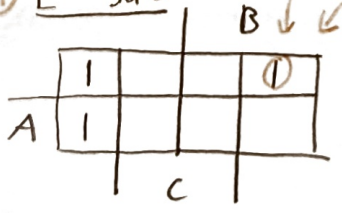
② B-Sa1



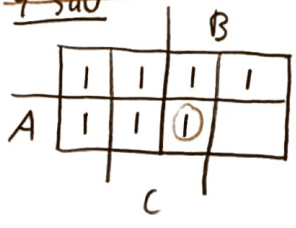
③ C-Sa0



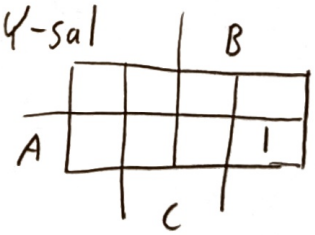
① ~~E-Sa0~~



③ ~~Y-Sa0~~



Y-Sa1



① A-Sa1 tests E-Sa0

② B-Sa1 tests E-Sa0

③ C-Sa0 tests Y-Sa0

* Ysa0 vs Sa1 flips w/ NOR vs OR

Final Fault List: A-Sa1, B-Sa1, C-Sa0, Y-Sa1

Collapse Ratio: 4/12 = .33